

LEGACY MIGRATION RISK ASSISTANT

IBM BOB 2.0 HACKATHON | lablab.ai | BUG SQUASH

LEGACY SQL SERVER

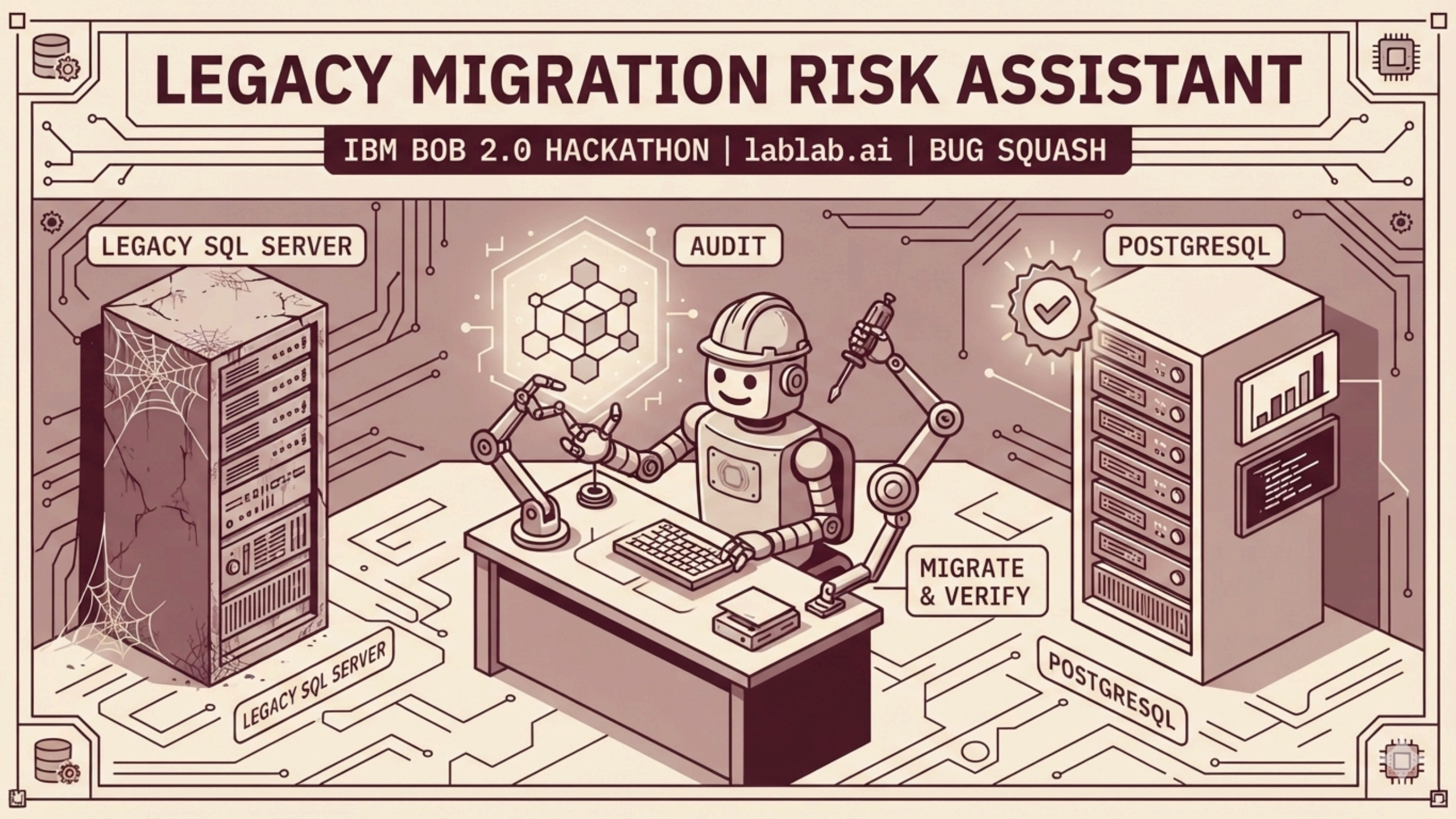
AUDIT

POSTGRESQL

MIGRATE
& VERIFY

LEGACY SQL SERVER

POSTGRESQL



Every Team Inherits a Database Nobody Wants to Touch

Legacy modernization is high-risk, low-glamour, and usually pushed to "next quarter"

- ▶ A real, widespread enterprise challenge — not a synthetic demo. We used Microsoft's own official 1998-era Northwind & Pubs schemas, genuinely deprecated, still common in production estates today
- ▶ Deprecated syntax, undocumented business logic buried in stored procedures and triggers
- ▶ Manually auditing migration risk typically takes a senior engineer **a full working day, per database**
- ▶ Most modernization tools stop at a risk report — a to-do list, and nothing actually moved
- ▶ We wanted to close that gap: audit, score, **and actually migrate**

Legacy SQL Server → Bob → Postgres

One real, working pipeline — not a diagram of one

Legacy SQL Server

- Northwind (1998)
- Pubs (1998)
- 13 procs/triggers
- 16 views
- No docs, no tests



IBM Bob 2.0

- Deprecated-syntax scan
- Business-logic extraction
- Risk scoring w/ evidence
- DDL + data-copy generation



PostgreSQL

- 13 tables + 13 FKs created
- 3,308 rows migrated
- Row counts verified
- Exact match to source



Fully containerized

SQL Server 2019 + PostgreSQL 16, both Docker — reproducible, zero local install



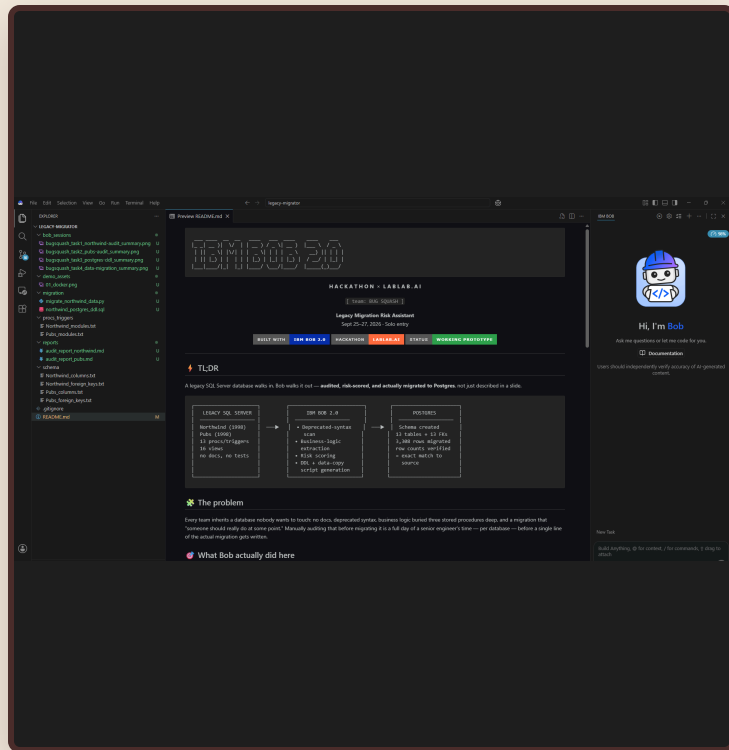
~160x productivity gain

8-hour manual audit → under 3 minutes, automated

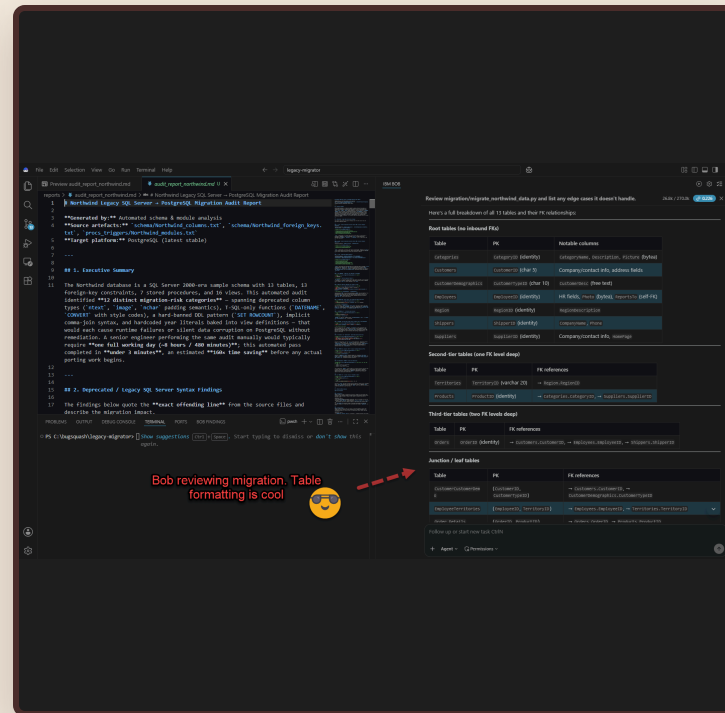
HOW BOB WAS USED

Four Distinct Jobs, One Real Pipeline

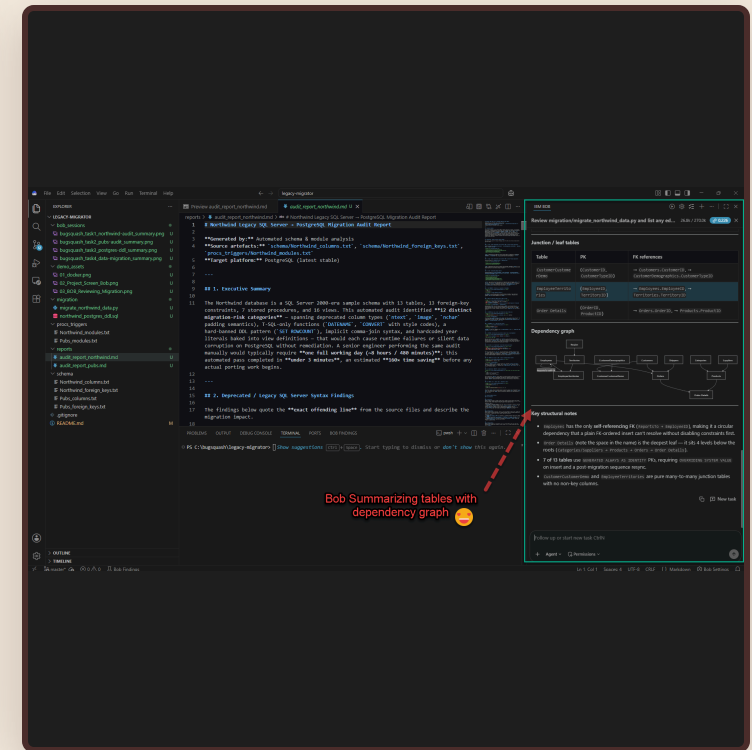
Audit · Generate DDL · Migrate · Verify



Bob IDE — project loaded, ready to work



Bob's dependency-graph analysis of Northwind's FK relationships

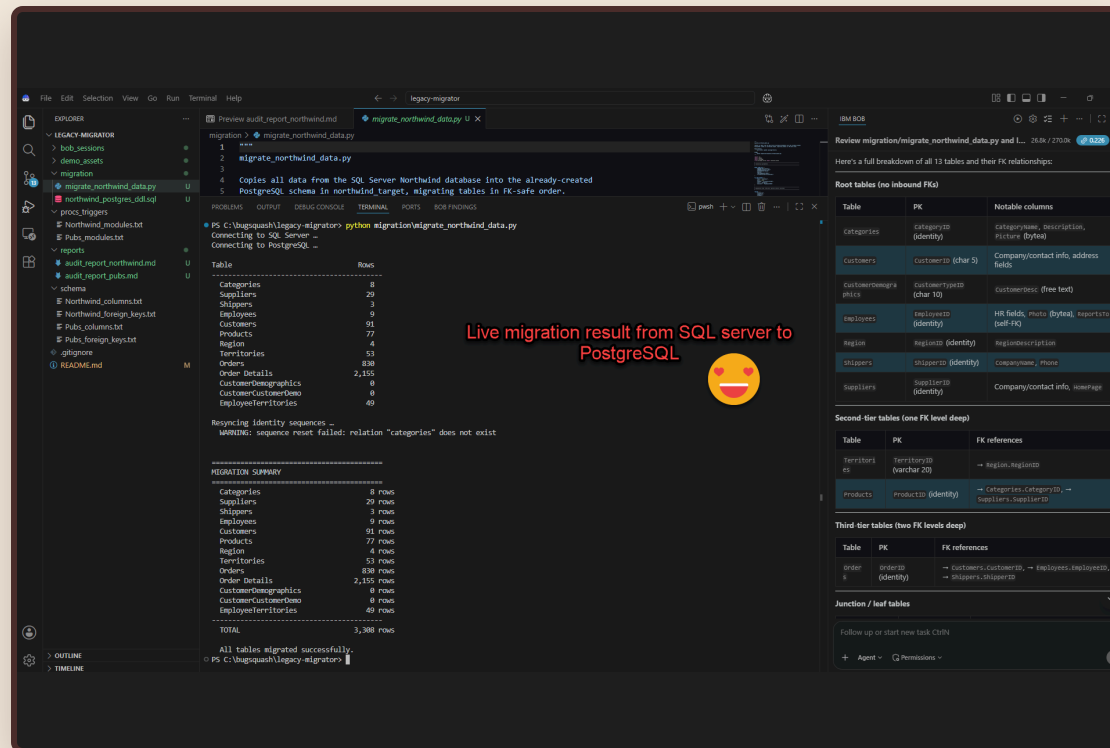


Bob executing live — reading files, running tool calls

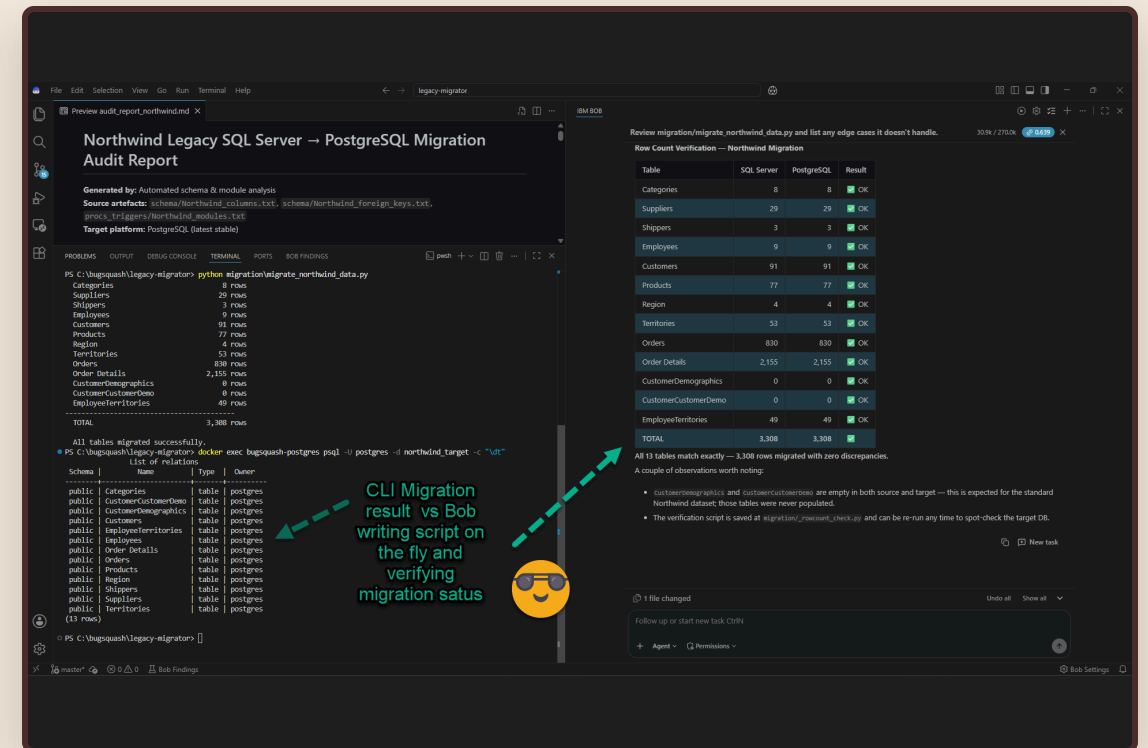
LIVE PROOF

Not Just Generated — Actually Executed

The migration ran, and Bob checked its own result



Migration script running — 3,308 rows moving into Postgres



Bob independently verifying the row counts match the source

Evidence-Based, Not Vague Warnings

Every risk score cites the exact offending line

Object	Type	Risk	Evidence
Ten Most Expensive Products	Proc	 5	SET ROWCOUNT 10 unsupported on Postgres
Invoices	View	 4	String concat + CONVERT (money , . . .)
Sales by Year	Proc	 3	DATENAME is T-SQL only
Employee Sales by Country	Proc	 2	Space-name only; otherwise ANSI
Alphabetical list of products	View	 1	SELECT *; space-name only

An 8-Hour Audit Became a 3-Minute Pass

And this time, the fix actually shipped

3,308

ROWS MIGRATED &
VERIFIED

13 / 13

TABLES & FOREIGN KEYS

~160x

FASTER THAN MANUAL
AUDIT

2x

VERIFIED — WIPED & RE-
RUN CLEAN

- Most legacy-audit tools stop at a report. This one closes the loop: Bob generated the migration, ran it, then independently verified its own result.

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github.com/viveknz/legacy-migrator

